

Nexilo - Architecture & Product Guide

1. Product Vision

Nexilo is an AI-powered platform that transforms PDF documents into actionable intelligence such as summaries, insights, structured data, and content.

2. Core Concept

User uploads a PDF → Processing pipeline extracts text → AI analyzes → Results are returned.

3. Architecture Strategy

Use a modular monolith architecture to ensure scalability while keeping development simple. Later evolve into microservices if needed.

4. Tech Stack

Java 21, Spring Boot 3, Maven, PostgreSQL, Angular, Docker, AI APIs (Google AI / OpenAI).

5. Package Responsibilities

user

Handles authentication, registration, roles, and user management.

document

Stores uploaded PDFs and metadata. No processing logic.

ai

Handles communication with AI providers, prompts, and response parsing.

processing

Core business logic. Orchestrates extraction, AI calls, and transformation.

subscription

Handles plans, limits, and billing logic.

common

Shared configs, security, exceptions, and utilities.

6. Processing Pipeline

Extract → Analyze → Transform → Store → Return result.

7. Best Practices

Use DTOs, async processing, clean architecture, and separation of concerns.

8. Growth Strategy

Start simple, iterate fast, then scale by splitting services (AI, processing, auth).

9. Key Success Factor

Focus on user experience, speed, and delivering real value from documents.